Correlation of Sunrise and Sunset Variations and Their Influence on Day Length

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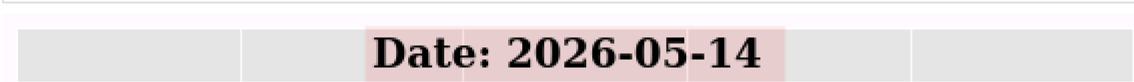
An exploratory analysis of sunrise and sunset timings from sunrise-and-sunset.com, focusing on their interrelationship and the resulting variations in day length.

In [137…

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
plt.style.use('ggplot')
%matplotlib inline
import matplotlib.dates as mdates
import seaborn as sns
import statsmodels.formula.api as smf
import datetime as dt

import plotly.io as pio
pio.renderers.default= 'notebook_connected'
import plotly.express as px
import plotly.graph_objects as go
```

In [138…

```
#This section of code display's today's date as a text in a matplotlib plot. Displaying i

fig,ax= plt.subplots(figsize=(10,0.5))
ax.text(
    0.5,0.5,  #set position
    "Date: "+str(pd.to_datetime(dt.datetime.today().date()))[:11],
    bbox=dict(facecolor='red', alpha=0.1), #make a red colored box around the text
    fontsize=20,fontweight="bold",family='serif', #format
    horizontalalignment='center',
    verticalalignment='center',
    )
ax.tick_params(  #remove axes ticks and labels
    axis='both',
    bottom=False, left=False,
    labelbottom=False, labelleft=False)
fig.set_facecolor(tuple(np.array([255, 250, 255]) / 255)) #set facecolor for the figure
```



In [139…

```
# Read data
url = "https://www.sunrise-and-sunset.com/en/sun" #Set url
df= pd.read_html(url)[0] #Read the dataset


#Cleaning
df= df.droplevel(1,axis=1) #drop unnecessary column levels
df=df.query("Sunrise!='midnight sun' & Sunset!='midnight sun'" )#Drop midnightsun cities
df= df.drop_duplicates("Country",keep="first") #drop duplicates

#Convert to Sunrise, Sunset and Day length column values to the datetime format
times= df.columns[2:]
for i in times:
    df[i]= pd.to_datetime(df[i], format="%H:%M")
```

In [140…

```
#Confirming data types and checking for missing values
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Int64Index: 224 entries, 0 to 226
Data columns (total 5 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   City        224 non-null    object
 1   Country     224 non-null    object
 2   Sunrise     224 non-null    datetime64[ns]
 3   Sunset      224 non-null    datetime64[ns]
 4   Day length  224 non-null    datetime64[ns]
dtypes: datetime64[ns](3), object(2)
memory usage: 10.5+ KB
```

Cities with the Shortest and Longest Days

The code below displays the top 5 cities with the shortest days and the last 5 with the longest days

In [141…

```
df.sort_values("Day length")[["City","Country","Day length"]]
```

Out[141…

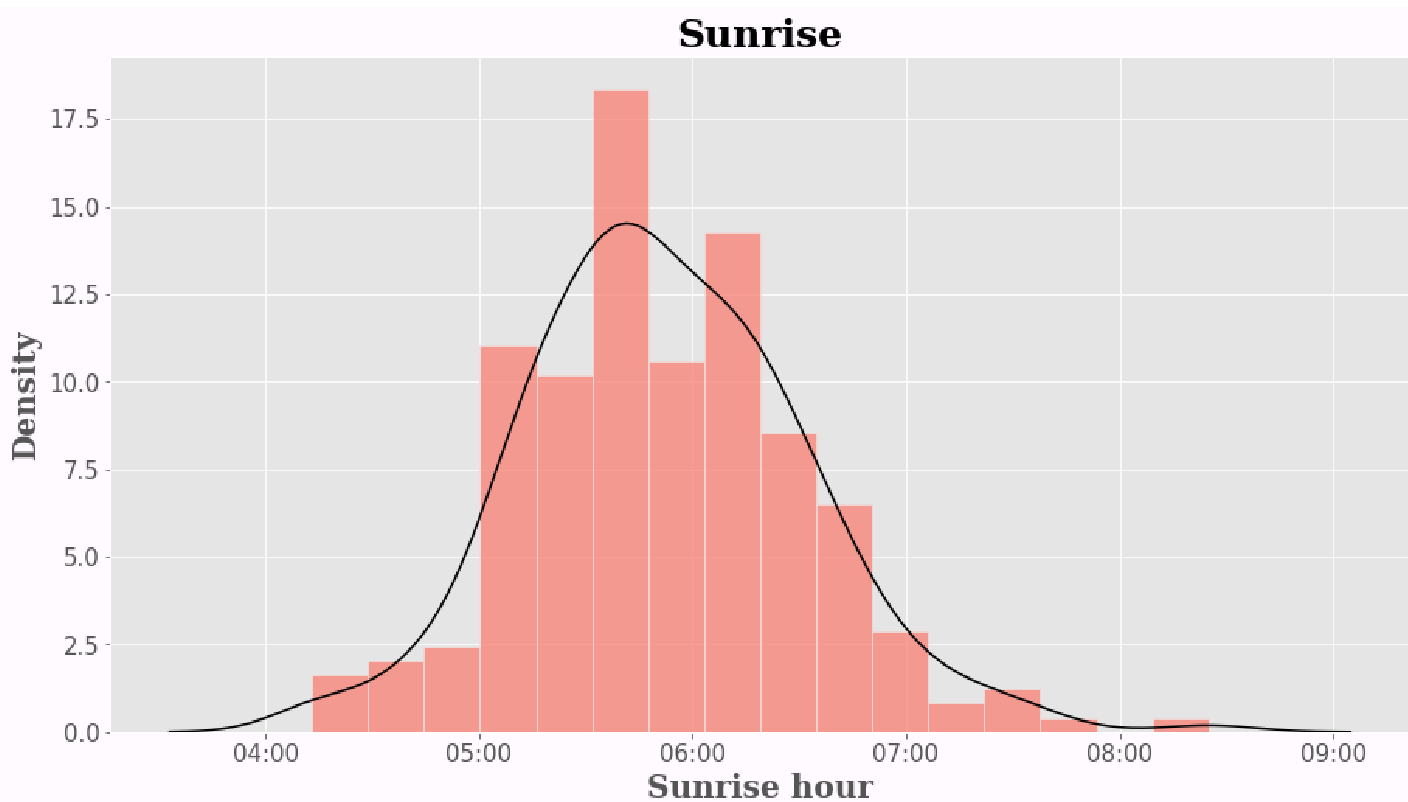
	City	Country	Day length
221	Stanley	Falkland Islands	1900-01-01 08:44:00
132	Wellington	New Zealand	1900-01-01 09:48:00
138	Canberra	Australia	1900-01-01 10:15:00
68	Montevideo	Uruguay	1900-01-01 10:16:00
0	Buenos Aires	Argentina	1900-01-01 10:17:00
...	...	...	...
207	Mariehamn	Åland	1900-01-01 17:07:00
117	Helsinki	Finland	1900-01-01 17:08:00
204	Tórshavn	Faroe Islands	1900-01-01 17:39:00
163	Reykjavík	Iceland	1900-01-01 18:21:00
201	Nuuk	Greenland	1900-01-01 18:22:00

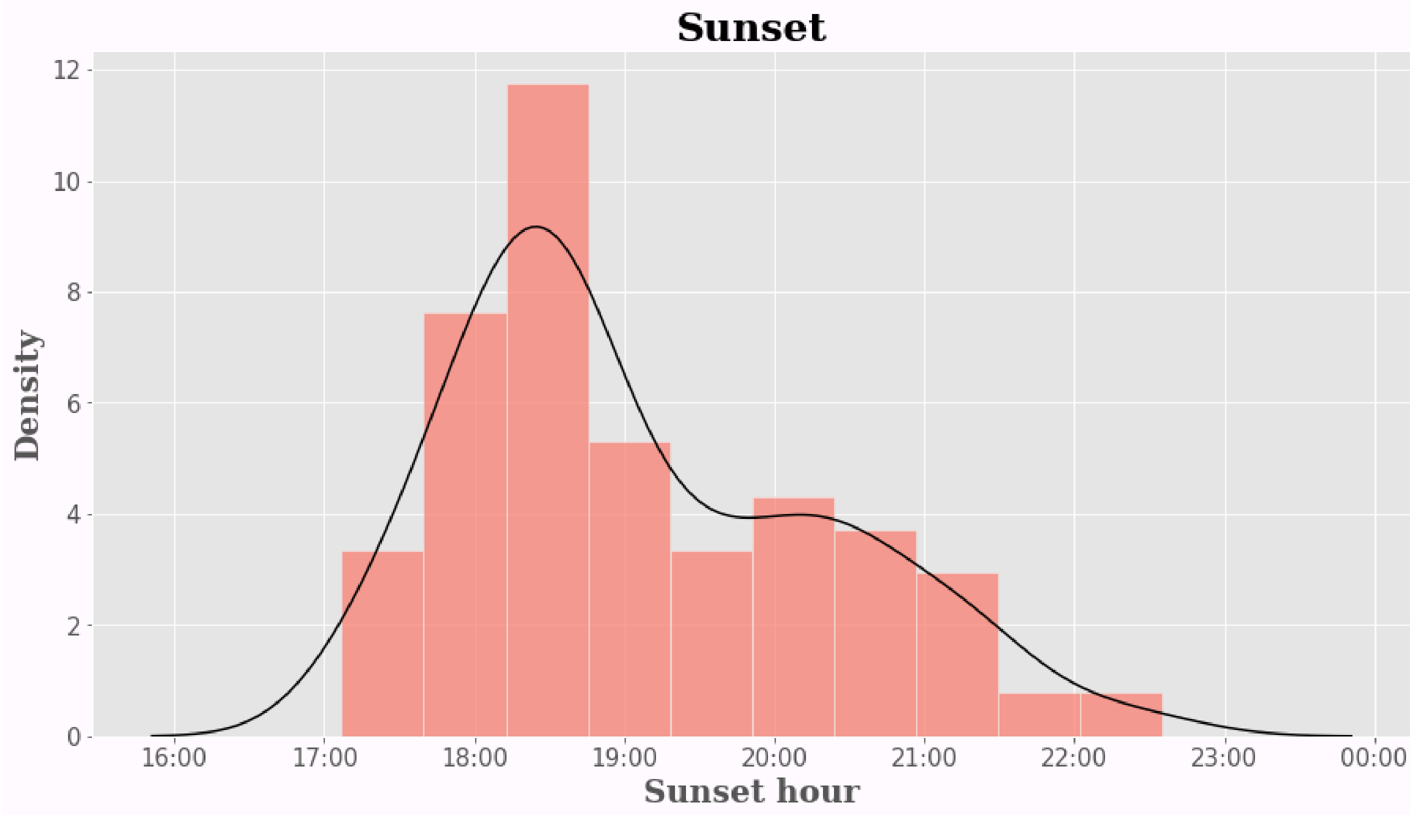
224 rows × 3 columns

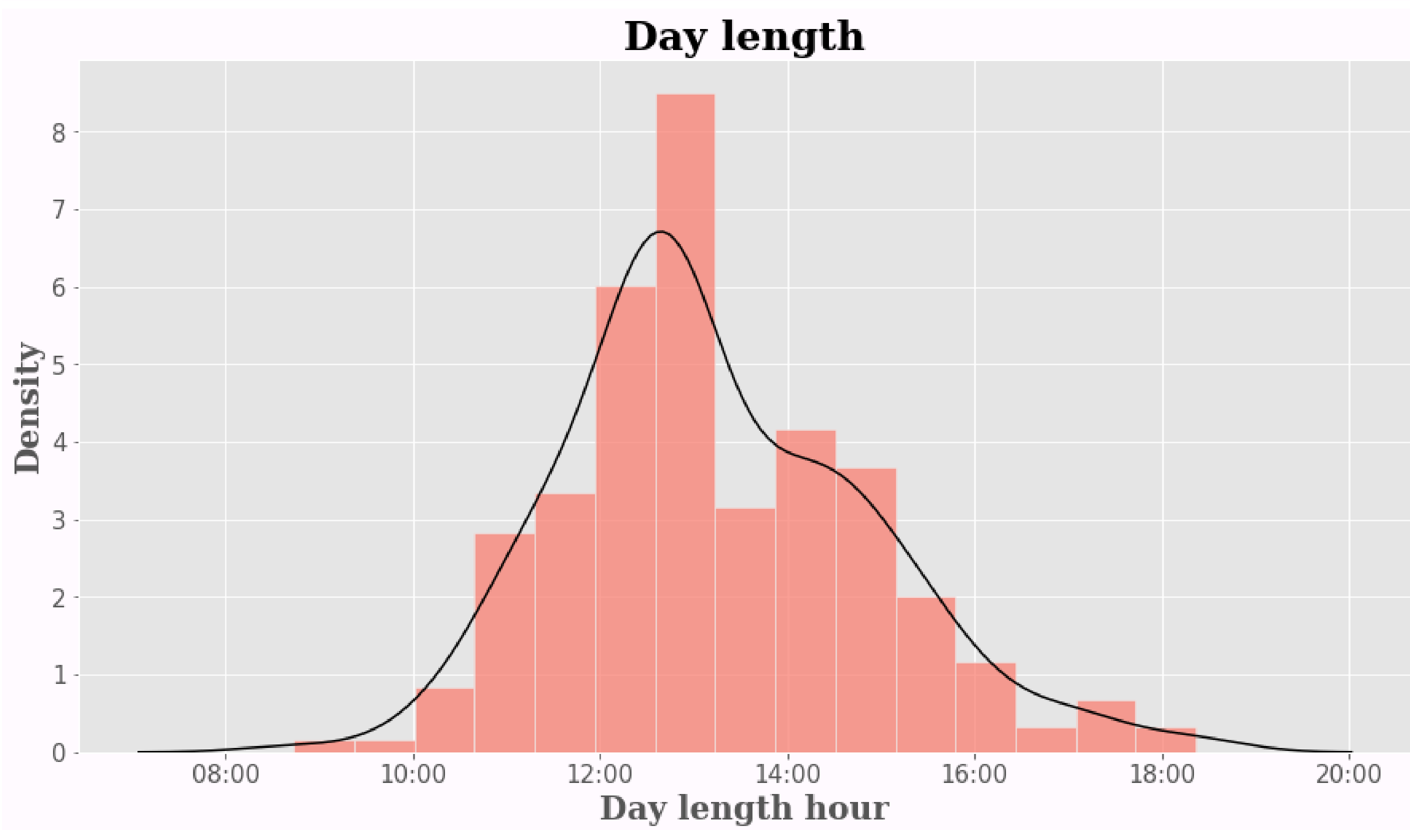
Density plots (Histogram + Kernel) for Sunrise, Sunset and Day Length

In [142…

```
def _hist(col:str):
    """
    This function plots a histogram and kernel density for a given series of data
    Input: series
    Output: Histogram and KDE plot
    """
    fig,ax= plt.subplots(figsize=(15,8))
    xformatter = mdates.DateFormatter('%H:%M') #format dates for axes labels


    sns.histplot(data=df, x=col,ax=ax,color='salmon',stat='density') #Plot histogram
    sns.kdeplot(data=df, x=col,ax=ax,color='k') #Plot Kernel Density

    #Format titles, labels and background color
    ax.set_title(col, loc='center', fontsize=25, fontweight="bold",family='serif')
    ax.xaxis.set_major_formatter(xformatter)
    ax.tick_params(axis='x', labelsize=15)
    ax.tick_params(axis='y', labelsize=15)
    ax.set_xlabel(col+" hour",fontsize=20, fontweight="bold",family='serif')
    ax.set_ylabel('Density',fontsize=20, fontweight="bold",family='serif')
    fig.set_facecolor(tuple(np.array([255, 250, 255]) / 255))

_hist("Sunrise")
_hist("Sunset")
_hist("Day length")
```







The visualizations above illustrate today's sunrise and sunset times, along with the total day length. It would be insightful to compare these results with the solar data from our respective current locations.

Scatter plot and Regression Line for Sunrise and Sunset

Let's explore how sunrise and sunset timings vary with each other

In [143…

```
#In this section of code, Daylength,Sunset and Sunrise datetime values are converted to se
# Sunset and Sunrise values need to converted seconds to run the regression
# Day length values need to be converted to seconds to plot the world maps in the end

def to_sec(col):
    """
    This function converts the datetime values to seconds

    Input: Series with time values in the datetime format
    Output: Series with time values in seconds
    """
    return col.dt.hour * 3600 + col.dt.minute * 60 + col.dt.second

#Make new columns for Day lengyh,Sunset and Sunrise which have values in seconds
df["Day l_sec"]= to_sec(df["Day length"])
df["Sunset_sec"]= to_sec(df["Sunset"])
df["Sunrise_sec"]= to_sec(df["Sunrise"])
```

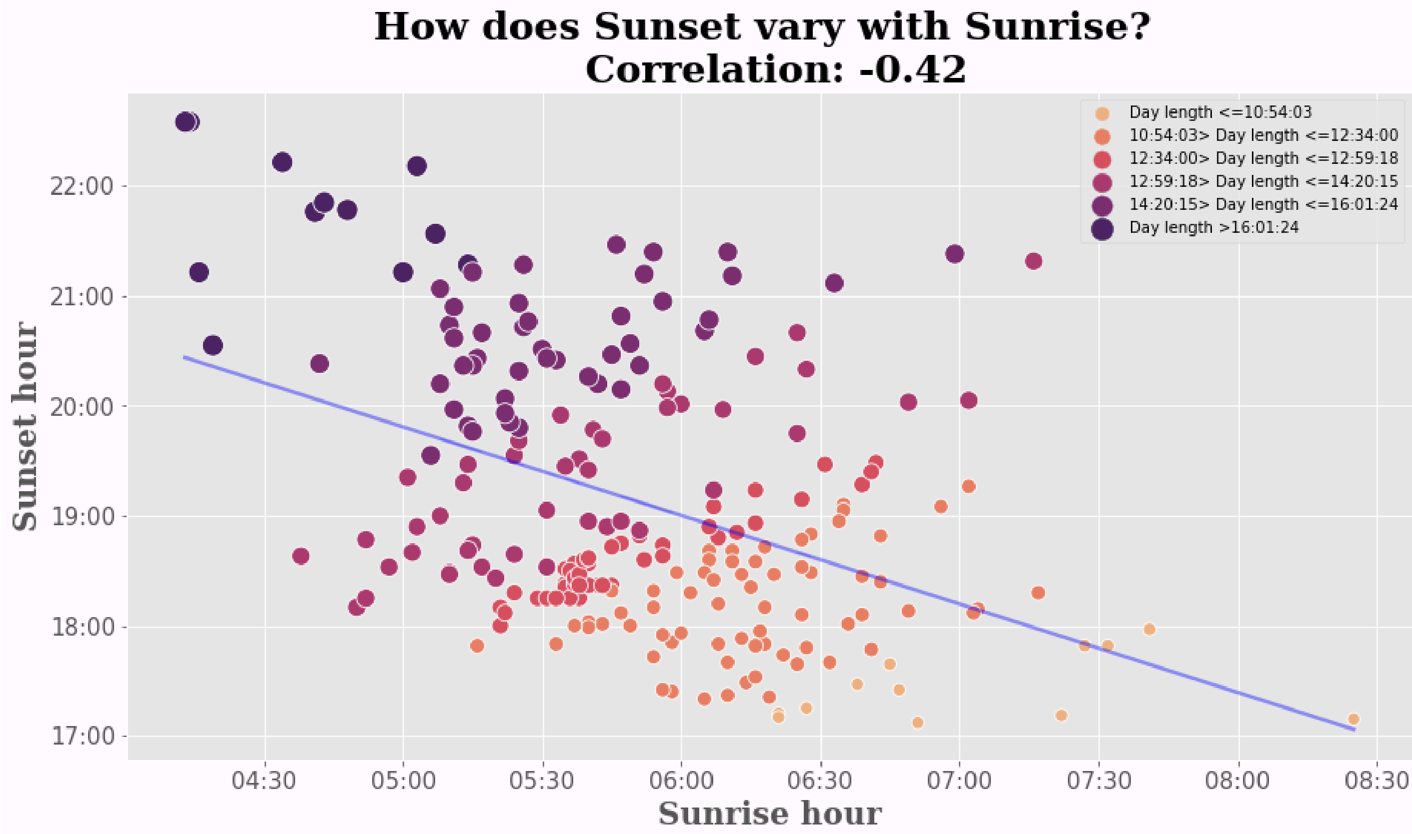
In [144…

```
#In this code block time from seconds is converted back to datetime values i.e the origina
#The purpose of this code block is to define a function such that it is able to convert th
# values back to datetime which then can be plotted on the same axes as the scatter plot.
# Remember that the scatterplot axes has datetime values not seconds.
# The predicted regression values are in seconds. These need to be convereted to datetime
#on the same axes

def to_dt(total_seconds):
    """
    This function convertes a value in total seconds to its respective datetime value
    Input: time in seconds
    Output: time in datetime
    """
    base_date = dt.datetime(1900, 1, 1)
    # Add the seconds to the base date
    result_time = base_date + dt.timedelta(seconds=total_seconds)
    return result_time

#Regress Sunset (in seconds) on Sunrise (in seconds) and store the preficted values in a c
#Convert the predicted seconds values to datetime using the to_dt function above

reg = smf.ols('Q("Sunset_sec") ~ Q("Sunrise_sec")',df).fit()
df['yhat'] = reg.predict()
df['yhat'] = df['yhat'].apply(to_dt)
```

In [145…

```
#Plot the scatter plot and regression

fig,ax= plt.subplots(figsize=(15,8))
xformatter = mdates.DateFormatter('%H:%M') #set format for axes labels

#Define bands depending on the magnitude of daylength according to which the points in the
#will vary in size and color

#Band 1: Bottom 5% of the Day length column values. Represented by a value of 1 in a colur
df["cols"]= np.where(df["Day length"]<=df["Day length"].quantile(0.05),1,6)
#Band 2: 5%-35%. Represented by a value of 2 in the cols column
df["cols"]= np.where((df["Day length"]>df["Day length"].quantile(0.05)) & (df["Day length"
#Band 3: 35%-55%. Represented by a value of 3 in the cols column
```

```
df["cols"]= np.where((df["Day length"]>df["Day length"].quantile(0.35)) & (df["Day length'
#Band 4: 55%-75%. Represented by a value of 4 in the cols column
df["cols"]= np.where((df["Day length"]>df["Day length"].quantile(0.55)) & (df["Day length'
#Band 5: 75%-95%. Represented by a value of 5 in the cols column
df["cols"]= np.where((df["Day length"]>df["Day length"].quantile(0.75)) & (df["Day length'
#Band 6: The remaining values are the Top 5%. They're set to the value of 6 in the cols co

#Plot scatterplot
sns.scatterplot(data=df,x="Sunrise",y="Sunset",hue="cols",size="cols",ax=ax,legend='full',


# The following function is for the legend labels
# Define a function which takes in quantile value and finds the associated Day-length valu
# corresponding to this quintile

def dayl_to_str(q):
    """
    This functions takes in a quantile values, finds the associated day length value and t
    converts it into a string value + removes extra charachters
    Input: Quantile e.g 0.05,0.35 etc
    Output: String value representine datetime
    """
    s= df["Day length"].quantile(q)
    s= str(s)[11:]
    return s

#Set new labels for the legend using the the function defined above
new_labels= ["Day length <="+dayl_to_str(0.05),
        dayl_to_str(0.05)+"> Day length <=" +dayl_to_str(0.35),
        dayl_to_str(0.35)+"> Day length <=" +dayl_to_str(0.55),
        dayl_to_str(0.55)+"> Day length <=" +dayl_to_str(0.75),
        dayl_to_str(0.75)+"> Day length <=" +dayl_to_str(0.95),
        "Day length >" +dayl_to_str(0.95)]



#Formal Legend
leg= ax.get_legend()
leg.set_title("")

leg.prop.set_size(10)
leg.prop.set_family('serif')

#Replace legend labels with new values
for t, l in zip(leg.texts, new_labels):
    t.set_text(l)

#Plot regression line
sns.lineplot(data=df,x="Sunrise",y="yhat",ax=ax, lw=2.4, color=(0,0,1,0.4),legend=False)


#Caclculate correlation to display in the title
cor= df[["Sunset_sec","Sunrise_sec"]].corr(method='pearson').iloc[0,1].round(2)
#Set title
ax.set_title("How does Sunset vary with Sunrise? \n Correlation: "+str(cor),loc='center',
#Formal axes
ax.xaxis.set_major_formatter(xformatter)
ax.yaxis.set_major_formatter(xformatter)

#Format axes ticks
ax.tick_params(axis='x', labelsize=15)
ax.tick_params(axis='y', labelsize=15)
#Format axes labels
ax.set_xlabel("Sunrise hour",fontsize=20, fontweight="bold",family='serif')
ax.set_ylabel('Sunset hour',fontsize=20, fontweight="bold",family='serif')

fig.set_facecolor(tuple(np.array([255, 250, 255]) / 255))
```



The analysis reveals a clear inverse relationship between sunrise and sunset times: later sunrises correlate with earlier sunsets, resulting in shorter day lengths.

In the visualization, cities marked in **deep purple** represent earlier sunrises and later sunsets (the longest days). Conversely, cities marked in **light orange** represent later sunrises and earlier sunsets, corresponding to the shortest day lengths in the dataset.

Plot choropleth world maps for Sunrise, Sunset and Daylengths

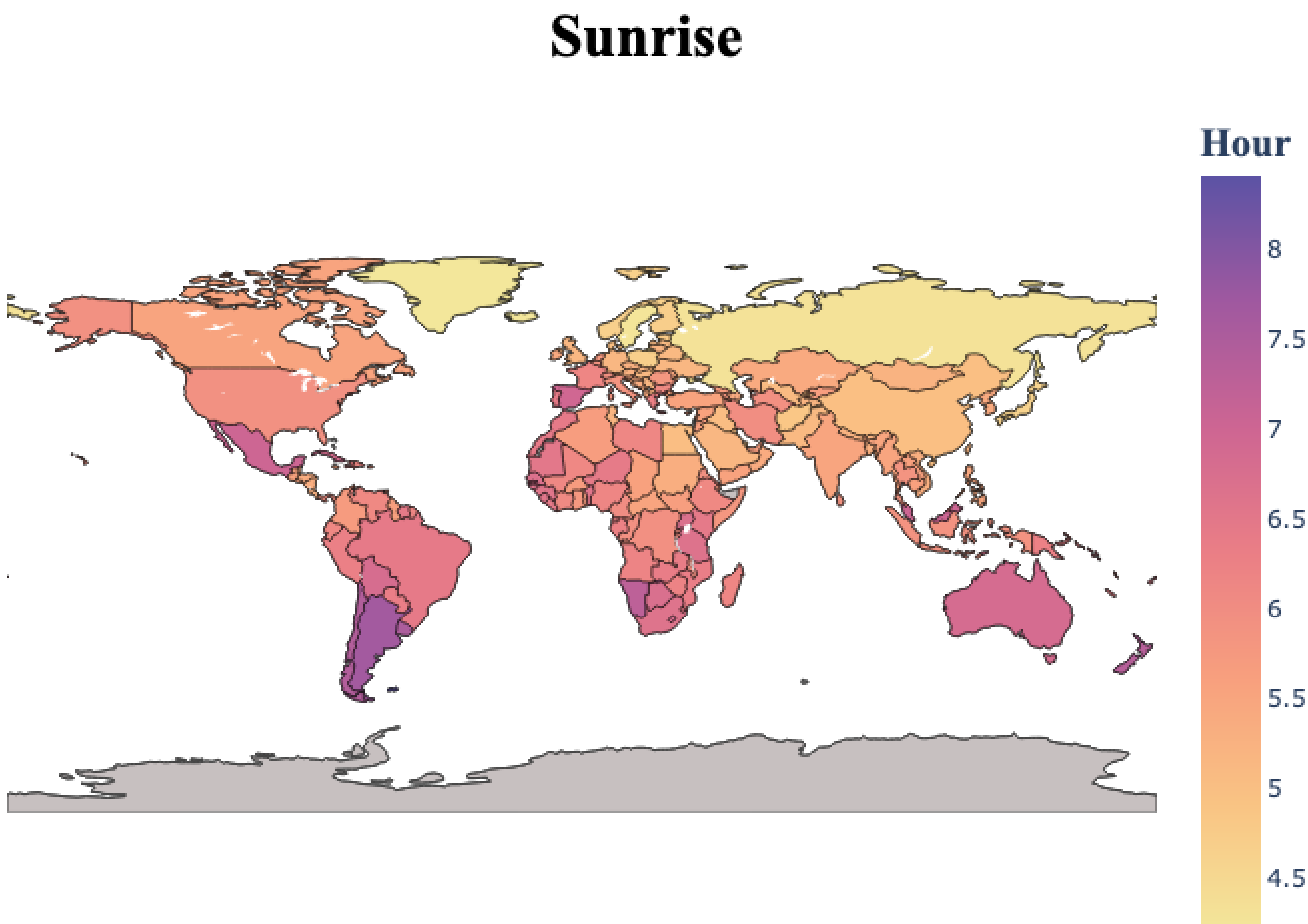
In [146…

```
#Merge iso3 country codes

ds = pd.read_csv('https://raw.githubusercontent.com/plotly/datasets/master/2014_world_gdp_
ds= ds.rename(columns={"COUNTRY":"Country"})

df_m= df.loc[:,["Country","Day l_sec","Sunset_sec","Sunrise_sec"]].merge(ds,on="Country",h

#Add codes for missing values
dicto={
    'South Korea':'KOR',
    'Congo-Kinshasa':'COD',
    'North Korea':'PRK',
    'Congo-Brazzaville':'COG',
    'Myanmar':'MMR',
    'Bahamas':'BHM',
    'Ivory Coast':'CIV',
    'East Timor':'TLS',
    'Reunion':"REU",
    'Curaçao':'CUR',
    'Cape Verde':'CPV',
    'Martinique':"MTQ",
    'French Guiana':"GUF",
    'Mayotte':",YT",
    'São Tomé and Príncipe':'STP',
    'Gambia':'GMB',
    'U.S. Virgin Islands':'VGB',
    'Guadeloupe':"GLP",
    'Åland':"ALA",
    'Saint Barthélemy':"BLM",
    'Micronesia':"FSM",
    'Turks and Caicos Islands':"TCA",
    'Falkland Islands':'FLK',
    'Wallis and Futuna':""}


for i in dicto.keys():
    df_m.loc[i,"CODE"]= dicto[i]

df_m= df_m.reset_index()

df_m= df_m.rename(columns={"Day l_sec":"Day length","Sunset_sec":"Sunset","Sunrise_sec":"S
```

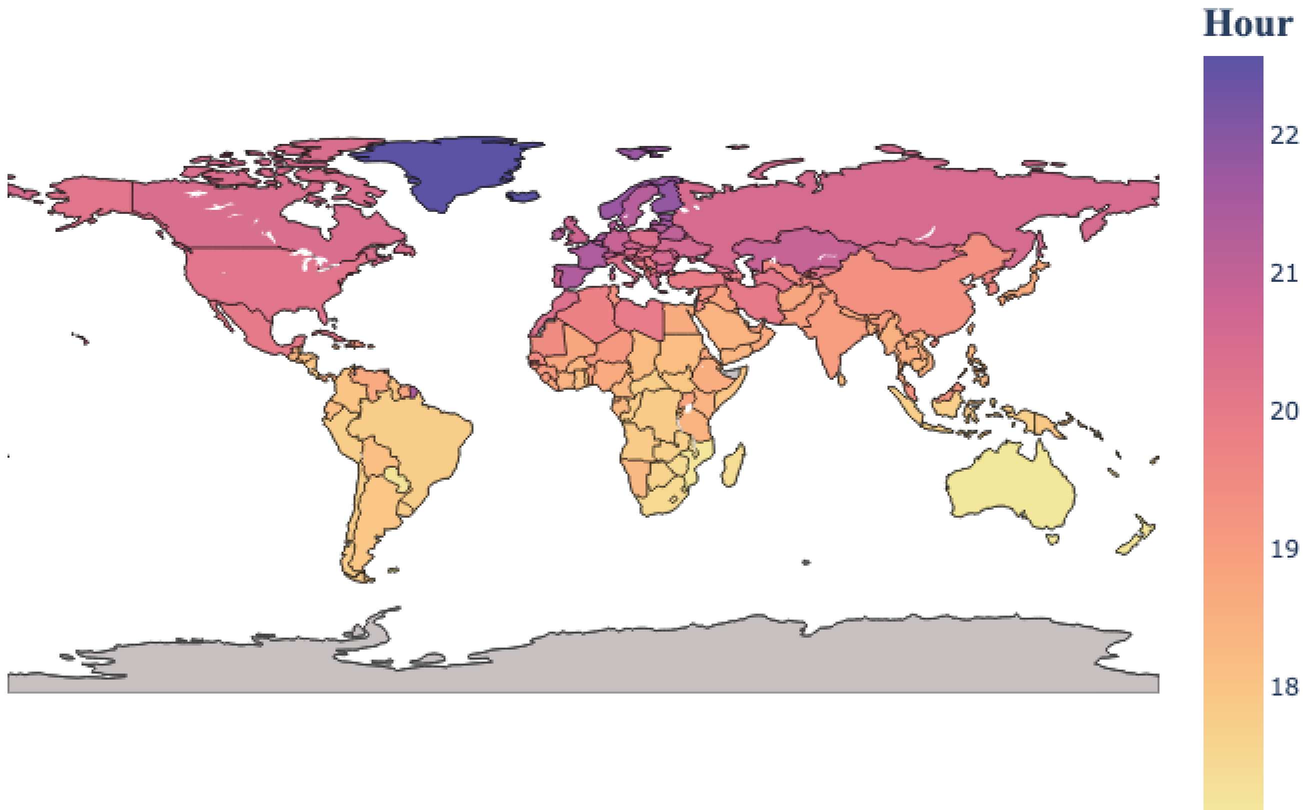
In [147…

```
#Plot maps

def map(col):
    """
    This function uses plotly to plot the maps but only outputs non-interactive images in
    Input: Relevant column whose data needs to be plotted
    Output: choropleth
    """
    fig = go.Figure(data=go.Choropleth(
        locations = df_m['CODE'], #locations identified based on country code
        z = (df_m[col]/3600).round(2),        #color varies based on magnitude of index
        text = df_m['Country'],
        colorscale = 'sunset_r',
        autocolorscale=False,
        reversescale=True,
        marker_line_color='black',
        marker_line_width=0.5,
        colorbar=dict(title={'text':'<b>Hour </b>','font':{'size':20,'family':'serif'}})
```

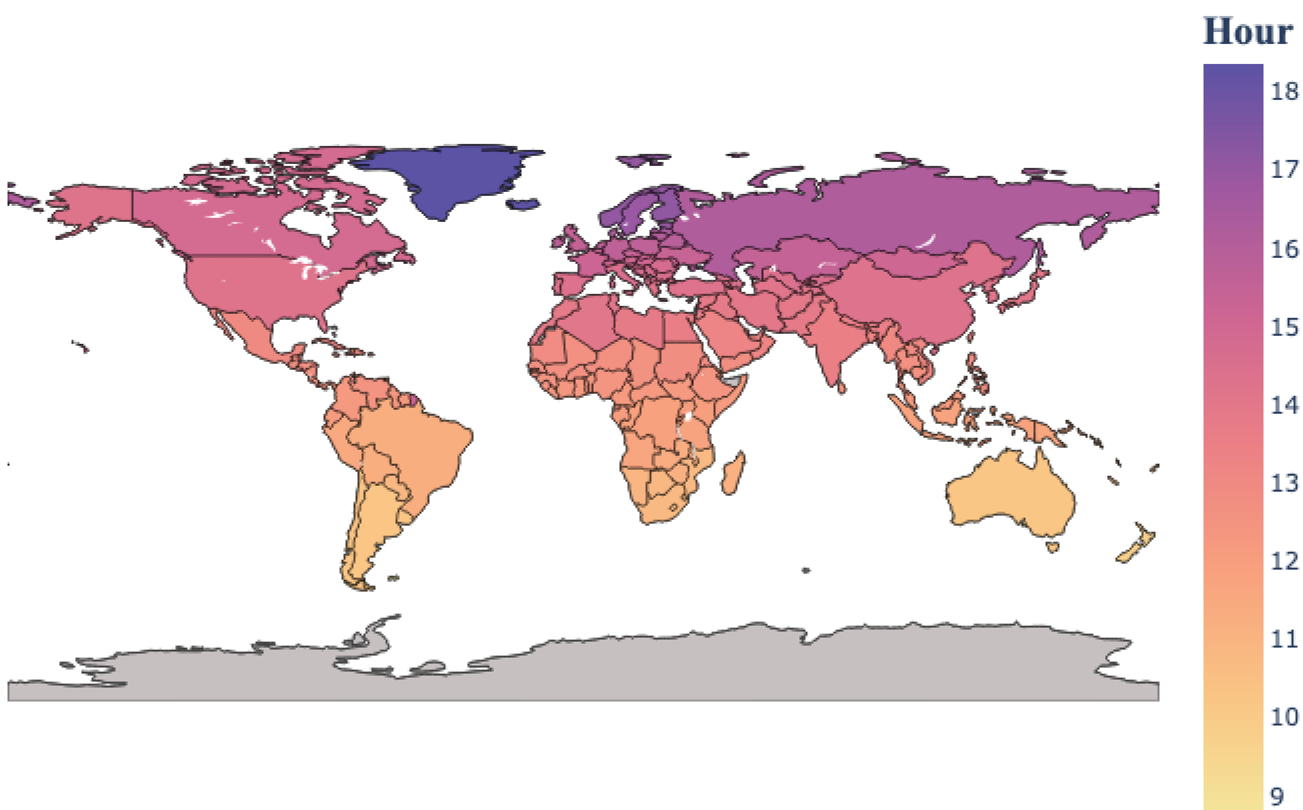
```
        ))

    fig.update_layout(
        title_text='<b>'+col,
        titlefont=dict(size =30, color='black', family='serif'),
        title_x=0.5,
        geo=dict(
        landcolor = 'rgba(175, 165, 166,0.7)', #customize grey color with missing values
        showcountries = True,
        showframe=False,
        showcoastlines=True,
        bgcolor='rgb(255,255,255)' #customize background color to our default value
        ),
        margin=dict(l=30, r=30, t=45, b=30),
        paper_bgcolor='rgb(255,255,255)',
    )

    fig.show("png")



map("Sunrise")
map("Sunset")
map("Day length")
```



Sunset



Day length



The maps above illustrate the global distribution of sunrise, sunset, and day length. How does the data for your country compare to these global trends?

In [ ]:

```
#Convert to pdf
! jupyter nbconvert --to webpdf Sunsets_and_sunrises.ipynb
```